

Module 1 — Introduction to Python Programming

Technical Notes

Course PYAIEC | Beginner Track | Duration: 8 hours

1. What is Python?

Python is a **high-level, interpreted, general-purpose** programming language created by **Guido van Rossum** and first released in **1991**. The name derives from the British comedy group *Monty Python's Flying Circus*, not the reptile. Python emphasises **code readability** and developer productivity, encoded informally in *The Zen of Python* (accessible via `import this`).

Core language characteristics

- **Interpreted:** source code is executed line-by-line by the Python interpreter (the reference implementation is **CPython**, written in C). No separate compilation step to machine code is required.
- **Dynamically typed:** variable types are determined at runtime, not declared explicitly.
- **Strongly typed:** implicit, unsafe type coercions (e.g. adding `int` to `str`) raise errors.
- **Multi-paradigm:** supports procedural, object-oriented, and functional programming styles.
- **Automatic memory management:** objects are reclaimed by a garbage collector (reference counting plus a cyclic collector).
- **Indentation-significant:** whitespace defines code blocks instead of braces, enforcing consistent layout.

Compiled vs. Interpreted

Python source (`.py`) is first compiled to platform-independent **bytecode** (`.pyc`), which is then executed by the **Python Virtual Machine (PVM)**. This hybrid model trades raw execution speed for portability and rapid development.

2. Why Python is Popular & Where It Is Used

Python’s adoption is driven by a gentle learning curve, a comprehensive **standard library** (“batteries included”), and an enormous third-party ecosystem distributed through the **Python Package Index (PyPI)**.

Domain	Representative tools / frameworks
Web development	Django, Flask, FastAPI
Data science & analytics	NumPy, pandas, Matplotlib
AI / Machine learning	TensorFlow, PyTorch, scikit-learn
Automation & scripting	<code>os</code> , <code>subprocess</code> , Selenium
Scientific computing	SciPy, SymPy, Jupyter
DevOps / Cloud	Ansible, Boto3 (AWS SDK)

3. How Python Powers Artificial Intelligence

Python is the de-facto language of AI and data science. Although Python itself is comparatively slow, the performance-critical numerical kernels of its AI libraries are implemented in **C, C++, and CUDA**, with Python serving as a high-level orchestration layer. This “glue language” role gives developers C-level speed with Python-level ergonomics.

Foundational AI/ML libraries

- **TensorFlow** — an end-to-end deep-learning framework from Google; builds and trains neural networks with GPU/TPU acceleration and supports production deployment.
- **PyTorch** — a deep-learning framework from Meta favoured in research for its dynamic computation

graph and Pythonic, imperative API.

- **scikit-learn** — a classical machine-learning library offering ready-made algorithms for classification, regression, clustering, and preprocessing; ideal for beginners.
- **NumPy / pandas** — the numerical and tabular-data backbone underpinning nearly all AI workflows.

Why researchers prefer Python

Concise, readable syntax shortens the experiment loop; the vast library ecosystem removes boilerplate; and a large open-source community accelerates problem-solving and reproducibility.

4. Setting Up the Python Environment

Installation

- **Windows:** download the Python 3.x installer from python.org; during setup, enable “Add python.exe to PATH” so the interpreter is callable from any terminal.
- **macOS:** download the official installer from python.org, or install via Homebrew with `brew install python`. (The system Python on macOS should not be used for development.)
- **Verify** the installation from a terminal / Command Prompt:

```
python --version      # Windows
python3 --version     # macOS / Linux
pip --version         # package installer
```

Development environments

- **IDLE** — the lightweight Integrated Development and Learning Environment bundled with CPython. It provides an interactive shell and a basic editor with syntax highlighting; suitable for first programs.
- **VS Code** — a free, extensible editor from Microsoft. With the official **Python extension** it adds IntelliSense (auto-completion), linting, debugging, and an integrated terminal — the recommended editor for ongoing work.

5. Execution Modes: Interactive vs. Scripting

Python code can be run in two distinct modes.

Interactive mode (REPL)

The interpreter runs as a **Read–Eval–Print Loop**: it reads a single statement, evaluates it, prints the result, and repeats. It is launched by typing `python` (or `python3`) with no arguments and is identified by the `>>>` prompt. Interactive mode is ideal for quick experiments, testing expressions, and exploring APIs; however, the code is not persisted.

```
>>> 2 + 3 * 4
14
>>> name = "Charan"
>>> print(f"Hello, {name}!")
Hello, Charan!
```

Scripting mode

Code is written to a `.py` source file and executed in full as a program. This mode is used for any non-trivial, reusable, or shareable program.

```
python hello.py      # run the saved script
```

	Interactive Mode	Scripting Mode
Input unit	One statement at a time	A complete <code>.py</code> file
Feedback	Immediate (per line)	After the whole script runs
Persistence	✗ Not saved	✓ Saved to disk
Best for	Experiments, testing	Real, reusable programs

6. Your First Python Program

The canonical first program prints a fixed message to standard output using the built-in `print()` function. A **comment** begins with `#` and is ignored by the interpreter. User input is read as a string with the `input()` function.

Listing 1: hello.py — basic input and output

```

1 # hello.py - my first Python program
2 print("Hello, World!")           # print a fixed message
3
4 name = input("Enter your name: ") # read text from the user
5 print("Welcome,", name)          # print with a variable

```

- `print(...)` — writes its arguments to standard output, separated by spaces, followed by a newline.
- `input(prompt)` — displays the prompt and returns the user's entry as a **string** (`str`).
- `#` — marks the rest of the line as a comment; comments document intent and are not executed.

Summary — Python is a readable, interpreted, dynamically typed language that dominates AI and data science through its C-backed libraries. Beginners install CPython, write code in IDLE or VS Code, and run it either interactively (REPL) or as saved `.py` scripts.